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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 24

Icing

Important Notice

Information pertinent to this test procedure is contained in Sections 1, 2 and 3. Further, Appendix A is applicable for identifying the environmental tests performed.

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24.0 Icing

24.1 Purpose of the Test

These tests determine performance characteristics for equipment that must operate when exposed to icing conditions that would be encountered under conditions of rapid changes in temperature, altitude and humidity.

24.2 General

Three icing test procedures are specified according to the category for which the equipment is designed to be used and installed in the aircraft (see Subsection 24.3).

Note: The selection of icing category depends on equipment location in (or on) the aircraft and the type of icing conditions expected. These conditions must be considered by the equipment designer in evaluating these requirements, which are determined by the end application and use of the equipment. These tests generally apply to equipment mounted on external surfaces or in non-temperature controlled areas of the aircraft where rapid changes in temperature, altitude and humidity are generally encountered.

These procedures specify test methods for evaluating the effects of various icing conditions on the performance or aircraft equipment, namely:

a. The effects of external ice or frost adhering to it.

b. The effects of ice caused by freezing of water condensation or by re-freezing of melted ice.

c. The effects of ice build-up caused by direct water exposure.

24.3 Equipment Categories

The following categories cover the anticipated ice formation conditions generally encountered in aircraft.

Category A

This test is applicable to equipment mounted externally or in non-temperature-controlled areas of the aircraft, where ice or frost may form due to condensation when the equipment is cold soaked to extremely low temperatures and subsequently encounters humid air at above freezing temperatures.

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Category B

This test is applicable to equipment with moving parts where such movement could be prevented or impeded by ice formation, or where forces resulting from the expansion of ice could damage structural or functional components. The ice formed in or on the equipment results from condensation, freezing, melting and/or re-freezing and may progressively accumulate water or ice inside non-sealed enclosures.

Category C

This test is applicable to items mounted externally or in non-temperature-controlled areas where there is risk of accumulating free water, which could subsequently freeze on the cold surfaces of the equipment. The test is intended to examine the effects of a representative thickness of ice on the performance of the equipment or to determine the maximum thickness that can be permitted before de-icing action is necessary. The required thickness and distribution of ice and any requirement for progressive build-up of ice shall be defined by the APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

24.4 Test Procedure

24.4.1 General

Mount the equipment under test in a manner representative of the normal installation in the aircraft. Remove all non-representative contaminants, such as oil, grease and dirt, which would effect adhesion between ice and the surfaces of the equipment under test, before beginning the appropriate tests. Operation of equipment that generates heat shall be limited to only that period of time necessary to determine compliance. The steps described in the Category A and Category B procedures are illustrated in Figure 24-1 and Figure 24-2.

24.4.2 Category A

a. With the equipment not operating, stabilize the equipment temperature at the low ground survival temperature specified in Table 4-1 at ambient room pressure and humidity.

b. As quickly as practical, expose the equipment to an environment of 30 degrees C with a relative humidity of at least 95 percent. Monitor the surface temperature of the equipment.

c. Maintain the environment at 30 degrees C and a relative humidity of at least 95 percent until the surface temperature of the equipment reaches five degrees C. As quickly as practical, change the environment to the appropriate ground survival low temperature at ambient pressure and humidity.

d. Repeat steps a. through c. for two additional cycles (total of three cycles).

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e. At the end of the third cycle, stabilize the equipment at the ground survival low temperature. Increase and maintain the chamber temperature to -10 degrees C and permit the surface temperature of the equipment to rise. When the surface temperature reaches -10 ±5 degrees C, place the equipment into the operating state and DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

Note: This test is designed to expose the equipment alternately to cold dry and warm moist environments. The use of separate chambers representing these two distinct environments is recommended.

24.4.3 Category B

a. With the equipment not operating, stabilize the equipment temperature at -20 degrees C at ambient room pressure. Maintain this temperature and decrease the chamber pressure to the appropriate maximum operating altitude specified in Table 4-1. Maintain this condition for a period of at least 10 minutes.

b. Raise the chamber temperature at a rate not exceeding three degrees C/minute while simultaneously increasing and maintaining the relative humidity in the test chamber to not less than 95 percent. Maintain this condition for a sufficient period of time to melt all frost and ice or until the surface temperature of the equipment reaches between zero to five degrees C. The chamber temperature should not be allowed to exceed 30 degrees C at any time during this step.

c. Increase the chamber pressure to room ambient at a uniform rate in a period of 15 to 30 minutes. At the completion of the re-pressurization, reduce the relative humidity in the chamber to normal room ambient.

d. Repeat steps a. through c. for a total of 25 cycles or as defined in the applicable equipment specification, whichever is less.

Note: If it becomes necessary to interrupt the test sequence, the interruption shall take place while the equipment is held at the low temperature condition.

e. At the end of the last test cycle, after the equipment temperature has stabilized at -20 degrees C, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

24.4.4 Category C

a. With the equipment not operating, stabilize the equipment at a temperature that will permit clear, hard ice to form on the equipment when sprayed with water.

Note 1: For this test, the ice formed shall be clear and hard (glaze ice). "White" or air-pocketed ice (rime ice) is not acceptable.

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Note 2: The optimum temperature is likely to be between -1 and -10°C depending on the thermal mass of the equipment and is best determined by experiment.

b. Build up a homogeneous layer of clear hard ice to a thickness defined by the APPLICABLE EQUIPMENT PERFORMANCE STANDARD by hand spraying a fine mist of water at a temperature that is close to freezing.

c. When the required thickness of ice has been achieved, discontinue the spraying. Place the equipment into the operating state and stabilize the equipment at a temperature of -20 degrees C. DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

Note: If multiple tests with increasing thickness of ice formation are required, a series of separate tests shall be performed with each thickness level formed in a continuous operation.

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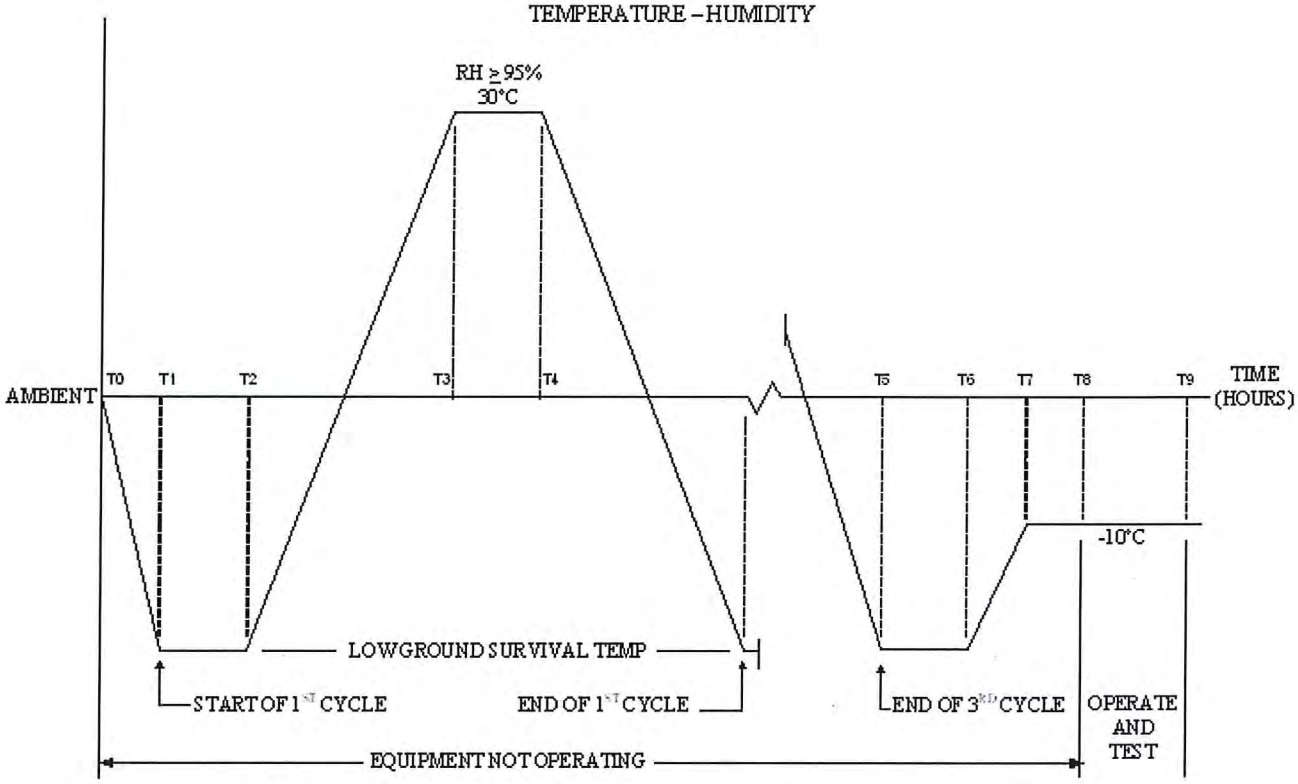

NOTES:

1. Unless otherwise specified, temperature and humidity change rates are optional.
2. T1 to T2 and T5 to T6 are times for equipment temperature to stabilize.
3. T2 to T3 and T4 to T5 are to be accomplished as quickly as practical.
4. T3 to T4 is time for surface temperature of equipment to reach 5°C.
5. T7 to T8 is time for surface temperature of equipment to reach -10°C.

FIGURE 24-1 CATEGORY A ICING TEST

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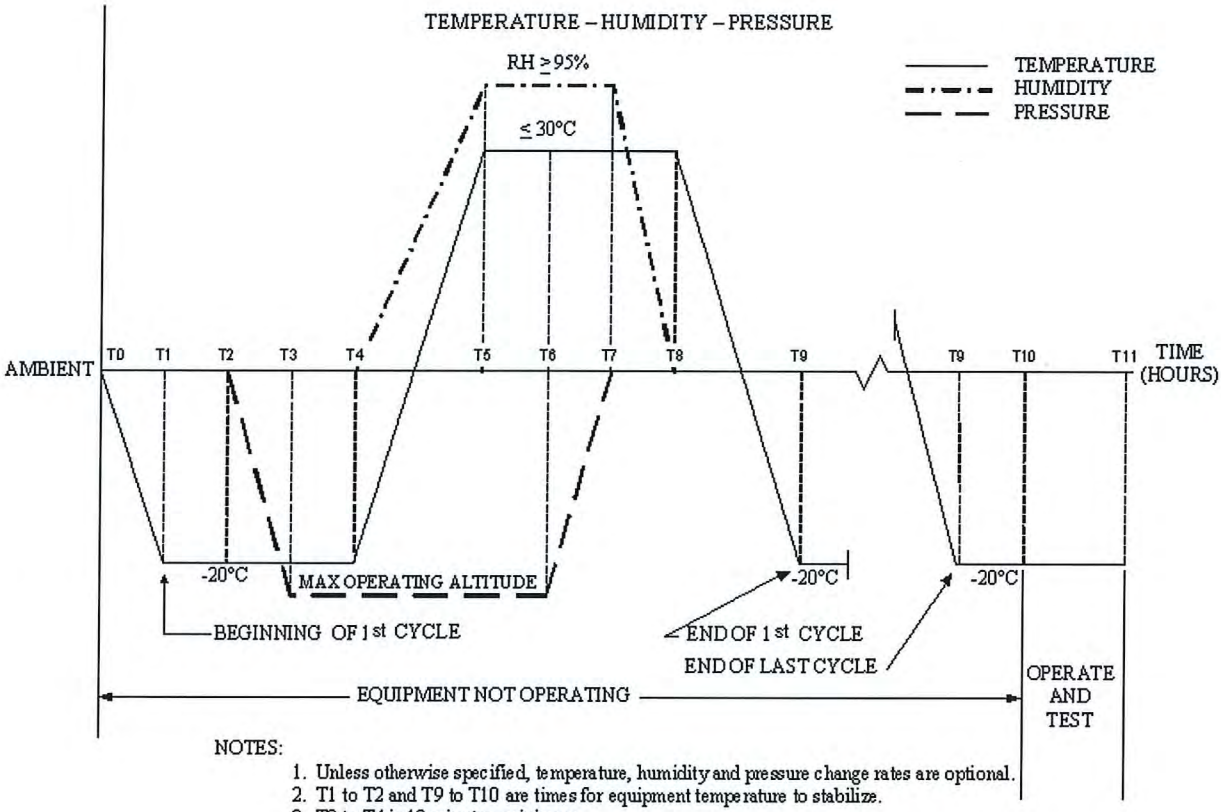

NOTES:

1. Unless otherwise specified, temperature, humidity and pressure change rates are optional.
2. T1 to T2 and T9 to T10 are times for equipment temperature to stabilize.
3. T3 to T4 is 10 minutes, minimum.
4. From T4 to T5 temperature rate of change is 3°C per minute maximum.
5. T5 to T6 is minimum time to melt all ice and frost.
6. From T5 to T8 chamber temperature should not exceed 30°C.
7. T6 to T7 is 15 to 30 minutes.

FIGURE 24-2 CATEGORY B ICING TEST

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